THE RELATION BETWEEN SIMULATED ARMYWORM DAMAGE AND CROP-LOSS IN MAIZE AND SORGHUM

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(Received for publication on 12th July, 1971)

The larvae of the armyworm *Spodoptera exempta* (Walk.) (Lepidoptera, Noctuidae) can inflict devastating damage to plants of the family Gramineae, including both wild and pasture grasses and also cultivated crops such as maize, sorghum millet, *Eleusine* (finger millet), rice, wheat and other cereals. When infestations on cultivated crops are very heavy, the plants may be damaged to the extent that little recovery takes place, and the whole crop may be lost. With lighter infestations regeneration may occur to varying extents.

The effect of attack on the yield in a cultivated grain crop is indirect since the damage is done to the vegetative parts of the plants, resulting in reduction of the photosynthetic area of the leaves. The extent of regeneration of the foliage depends on a number of factors, such as the age and manner of growth of the plant, and on variable and largely unpredictable factors such as rainfall and soil conditions. It is therefore not easy to assess the crop loss which will result from a given amount of leaf damage, and accordingly at what level of damage control can be applied to economic advantage.

The experiments here described were designed to obtain an estimate of the crop loss to be expected from different degrees of damage in the case of maize and sorghum millet, two of the staple food crops of East Africa.

Owing to the improbability of finding conveniently situated natural infestations of larvae, with a variation in density suitable to produce the required range in levels of leaf damage, and on cultivations available for crop estimation, it was decided to simulate damage artificially. This is one of the recognized ways of estimating the relation between insect damage and crop loss (Judenko, 1965); similar methods have been used with plants as diverse as green peas (George, Light and Gair, 1962), sugar beet (Jones, Dunning and Humphries, 1955), and pine trees (Austara, 1970). The damage was done by removing leaves, or portions of leaves, and even stems in plots of different ages arranged in a randomized block design, and measuring the resulting grain yield; in this way some broad conclusions were reached about the relation between damage-level and crop loss.

A number of earlier experiments of a similar nature on maize have been described (Dungan, 1930; Eldredge, 1935; Camery and Weber, 1953; Hanway, 1969); they largely relate, however, to defoliation in later stages of development of the plants than we are usually concerned with in armyworm damage, and trials concentrating on earlier stages therefore seemed to be required.

In 1969 a preliminary experiment was done with one variety of maize at the East African Agriculture and Forestry Research Organization (E.A.A.F.R.O.) at Muguga, near Narobi, Kenya. Although it provided some useful information it was unsatisfactory in certain respects. It was repeated under more favourable circumstances with two varieties of maize in 1970; in the same year a similar experiment was conducted with maize at Kitale, Kenya, with the co-operation of the Maize Genetics Research Division of E.A.A.F.R.O., and another with sorghum at Serere, Uganda, with the co-operation of the Sorghum Research Division of E.A.A.F.R.O. The account which follows incorporates all these experiments.

METHOD

(*a*) *The treatments.* In each plot all the plants were either left untreated as controls, or were subjected to one of three treatments:—

A—75 per cent leaf removal, effected by cutting off $\frac{3}{4}$ of the area of every individual leaf.

B—100 per cent leaf removal, or cutting off every leaf right to the base, leaving a bare stem.

C—Removal of the entire stem, together with the leaves, by cutting it at ground level.

Treatment A was done in two ways (Fig. 1): the first (*a*) consisted of removing with scissors a series of triangular pieces of leaf along either side of the midrib, without damaging the latter, and then removing every alternate triangular piece remaining; this is similar to a method

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used by Dungan (1930) who removed 4 in. segments alternately along both sides of the leaf. This method being time consuming, a simpler alternative (*b*) was used in most of the 1970 experiments, only involving cutting off the distal half of each leaf, and then stripping off the lamina on one side of the remaining basal half. In either case 25 per cent of the original leaf remains.

Since the extent of regeneration is known to depend upon the stage of growth of the plant when the damage is done, each treatment was applied at three different stages, the stage of development being better indicated by the number of opened leaves than by age. In the 1969 experiment on maize at Muguga, the stages selected were:—

(i) 4-5 leaves open.
(ii) 8-10 leaves open.
(iii) 13-15 leaves open.

These stages occurred at approximately 18, 42 and 69 days after planting.

In the 1970 maize experiments at both Muguga and Kitale, these stages were somewhat advanced:—

(i) 3-4 leaves open.
(ii) 6-8 leaves open.
(iii) 10-12 leaves open.

These stages occurred at 19, 30 and 46 days after planting at Kitale, and 26, 46 and 61 days after planting at Muguga; at the latter this was about 11, 31 and 46 days after germination.

The reason for the change will become apparent later when the behaviour of the growing point is described. From knowledge of the latter, and from experience gained in the 1969 experiment, it was concluded that treatment C. 3 (stem and leaves cut off in the latest stage) would inevitably result in complete destruction of all the maize plants; this treatment was therefore omitted in the 1970 experiments, and the plots set aside for it therefore became available as additional controls.

In the sorghum experiment at Serere, treatments (i), (ii) and (iii) were made at the 4-5, 8-9 and 12-14 leaf stage respectively (about 17, 31 and 40 days after planting). The plants started to produce tillers at the five to six leaf stage, and the leaf count was made on the main stem.

(*b*) *The varieties of maize and sorghum used.* In the 1969 trial at Muguga the maize variety "Katumani Composite B" was used, for this altitude (6,800 ft.) the most nearly suitable one available. In 1970 at Muguga the same variety was used again, and also Hybrid 512; these are respectively "short-term" and "long-term" varieties, with shorter and longer development periods; at Kitale, the two hybrids 512 and 611C were used, the latter growing somewhat taller and taking a little longer in development. These combinations enabled a comparison to be made between varieties with different development periods, and between results for one variety grown in two different localities.

At Serere the sorghum variety "Serena", a popular one widely grown in Uganda, was used.

(*c*) *Design of experiments.* The same basic design was used in all the experiments, a randomized block design with four replications. In each block two plots were used as controls, and one for each of the nine combinations of treatments and stages of growth, making 11 in all. In two experiments a twelfth plot was included in each block as a spare, but was not used.

Each plot contained four rows, the two inside rows only being used for assessments and the two outer ones as guard rows to minimize the border effect. The detailed arrangement within plots in the different experiments was as follows:

	No. of plants per row	Distance between rows	Distance between plants	Maximum No. of plants measured per plot
Maize				
Muguga, 1969	13	86 cm.	25 cm	26
Muguga, 1970	18	80 cm.	26⅔ cm	36
Kitale, 1970 . . .	18	80 cm.	26⅔ cm	36
Sorghum				
Serere, 1970	16	2 ft. (= 61 cm.)	6 ins. (= 15 cm.)	32

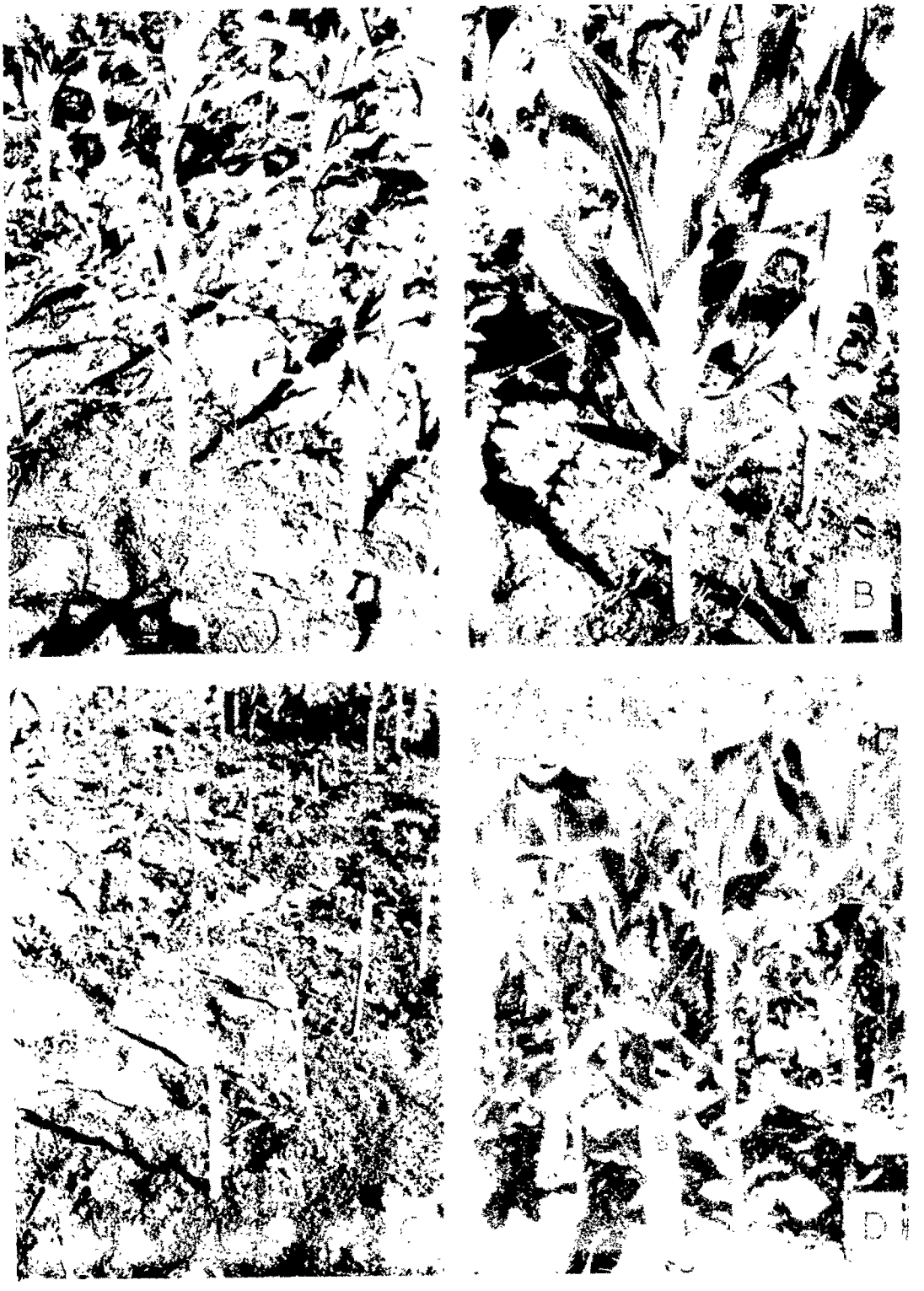

Plate 1—Simulated armyworm damage to maize

A. Treatment A.2. 75 per cent leaf removal; 1 day after treatment.

B. Treatment A.2. 26 days after treatment.

C. Treatment B.2. 100 per cent leaf removal; immediately after treatment.

D. Treatment B.2. 34 days after treatment

Two seeds were planted at each hill; at the time of the first treatment these were thinned to one per hill, the most healthy plant being retained.

(*d*) *Weeding, fertilizer application and pest control.* Weeding was carried out when necessary; in the Muguga experiments this amounted to three times during the earlier stages of growth.

No fertilizer was used in the 1969 Muguga experiments, so that the local soil variations which were eventually found to exist were not evened out. In the 1970 Muguga and Kitale experiments single superphosphate was spooned into the planting holes and mixed with the soil at the time of planting at the rate of 250 kg./ha. and a nitrogenous fertilizer was applied at approximately the eight to nine leaf stage. At Serere, 200 lb. of calcium nitrate and 200 lb. of single superphosphate per acre were applied.

Insect pests were a sufficient threat to require treatment. In the 1969 Muguga experiment there was damage by cutworms; spraying with dieldrin (Kynadrin) caused scorching of the leaves, severe enough to necessitate replanting, which resulted in a setback to development; a bran-molasses-aldrin bait was subsequently used and was more successful. In the 1970 Muguga experiment the appearance of aphids threatened to cause the spread of virus disease, and the aphids were treated with Rogor 40 per cent at a rate of 10 fluid oz./acre on May 8/9th, about six weeks after planting; a second application was made on June 10th.

To control shoot-fly on the sorghum at Serere, endosulfan was applied at the rate of 1 lb./acre, at about the four to five leaf stage and again about 10 days later.

Birds (seed-eating species of the family Fringillidae) constituted a serious threat to maize cobs in the Muguga experiments. In 1969, the damage was considerable and no adequate protective measures were taken, so that it was necessary to make a correction for damaged cobs in estimating yield. In 1970, bird-scarers were employed during the hours of daylight for the whole period between cob formation and harvest, and damage was negligible.

Virus diseases became established in some of the maize plants at Muguga in 1970, and are dscussed elsewhere; no control measures were used other than that against aphids as potential carriers, mentioned above.

In the maize at Muguga in 1970 a few cobs were found in early August with a rot at the base, in some cases extending up the cob and into adjacent leaves. This was identified as "pink ear-rot" (*Fusarium graminearum*), and was probably encouraged by wet weather; nothing could be done about it at this stage, but very few plants were involved and effects on plot yields were negligible.

Data Recorded

Although the ultimate yield of grain in relation to treatment was the parameter of of greatest importance, a number of other measurements and records were taken at appropriate periods during the growth of the plants, as a basis for assessing and comparing their regeneration potential after different treatments. The measurements, although not all recorded in all experiments, were taken as follows:—

Yield. The total grain for the middle two rows of each plot was removed from the sun-dried cobs or heads and weighed. The moisture content was then estimated with a "supermatic" moisture meter (or with a "cera" moisture tester in the case of samples less than 250 gm. in weight), and the weight corrected to what it would have been at the standard 13 per cent moisture content.

In the Kitale maize experiment, largely owing to excessive rain, a few cobs suffered from rot, and the grain was so reduced that it was unusable; the yield was estimated on "usable" cobs only. In the Muguga 1970 maize experiments a number of plants became infected with two virus diseases (Maize Streak and Sugar-cane Mosaic), and the grain from them was weighed separately; although this had a significant effect on the yield of individual plants, the incidence of virus was sporadic and not significantly related to the treatments except in the Katumani maize, where the mosaic virus was slightly but significantly greater in control than in treated plots; the yield from control plots would therefore have been actually and relatively a little greater without the virus. This factor was insufficient to affect general conclusions, and no correction was made for it. Some of the grain samples from plants affected by virus weighed less than 100 gm. and the moisture content could not be estimated in either supermatic or cera moisture meters; in such cases it was taken to be the

same as that of the grain from healthy plants in the same plots, since it had been similarly treated.

Number of cobs or heads per plot. The total number of cobs (maize) or heads (sorghum) produced in each plot were counted. In the 1970 maize experiment at Kitale the few diseased and rotted cobs were included, since they were formed after the treatments and there was no indication that rotting resulted from particular treatments; for similar reasons, the cobs from virus-infected plants were included together with those from healthy plants in the 1970 Muguga experiment, and the "undeveloped" cobs (those in which no grain was formed) in the 1969 Muguga experiment.

Weight of grain per cob or head. This was assessed only on the "usable" cobs in the Kitale experiment, and on the healthy cobs (from plants unaffected by virus) in the 1970 Muguga experiment. In the 1969 Muguga experiment it was assessed on all cobs, both developed and undeveloped, since the latter condition may be expected to be related to the treatment rather than to extraneous factors such as disease.

Plants without cobs, and with undeveloped cobs. Under the poor growing conditions at Muguga in 1969 the plants suffered from reduced powers of regeneration, and the effects of treatment were more severe. Although most plants produce one cob each, on this occasion many produced none at all, and of those cobs produced a proportion (termed "undeveloped") did not carry any grain. These were counted, but this was not repeated in 1970 when growing conditions were better and the figures would have been much lower.

Height of plants. This was measured in cm. from ground level to the base of the topmost leaf, when the plants were full grown. The average was calculated for all plants in the middle two rows of each plot, and for all plots with each treatment. It would have been rather meaningless for sorghum because of the irregularity of tillering, and was difficult in the maize at Kitale owing to the excessive height and frequent lodging of the plants; it was therefore recorded only in the maize experiments at Muguga.

Rate of development. Since defoliation might be expected to affect the development of a plant, three indicator stages were recorded —tasselling, silking and cob formation. The number of days after planting was recorded when 50 per cent of the plants in the centre rows of each plot reached these stages; the average for all plots with each treatment was calculated. This was done only in some of the 1970 maize experiments at Muguga and Kitale.

Number of tillers (Sorghum only). The number of tillers in all plants was counted in the sorghum experiment at Serere, regardless of whether they bore heads of grain or not.

Number of plants surviving after "C" treatments. The number of plants surviving after removal of the complete stem and leaves at different stages of growth was of particular interest in relation to the behaviour of the growing point, as will be explained below. The number of plants surviving was counted in all four rows in all plots given "C" treatments.

Behaviour of the growing point in maize. The growing point, or primary meristem in maize is known to remain below ground level until the plant has developed several leaves, and it position when the stem is cut to ground level in "C" treatments is therefore important. Extra plots of the three maize varieties Katumani Composite "B", Hybrid 512 and Hybrid 611C were planted at Muguga in 1970; at different stages of growth, recorded as the number of opened leaves (counting a part-opened terminal leaf as $\frac{1}{2}$), plants of each variety were dug up, after marking the position of ground level on the stem. The plants were then slit down the middle of the stem and the position of the growing point with relation to ground level was measured, and recorded as − when below ground level and + when above.

RESULTS

The results are described below separately for the various data recorded, with the figures tabulated to show statistically significant differences (Tables I—VI). The major results are also all shown together in Table VII, where treatment results are expressed as percentages of the controls.

Yield

The average yields per plot for all treatments, and for all experiments, are given in Table I.

Taking first the maize experiments, there were no survivals in plots with C.3 treatments, and few in those with C.2, and the figures for

TABLE I—ARTIFICIAL DAMAGE EXPERIMENT: MEAN YIELDS FOR CONTROLS AND TREATMENTS IN GRAMS PER PLOT

In each experiment, figures with same letter appended are not significantly different from one another. Figures in brackets were not included in significance tests. Instances where treatments were followed by yields greater than the controls are marked with an asterisk[a].

Experiment	Control mean	Treatment Means									Least significant difference (5 per cent level)	
		A.1	A.2	A.3	B.1	B.2	B.3	C.1	C.2	C.3	Between treatment means	Between control and treatment means
MAIZE												
Katumani Composite "B"												
Muguga 1969	615[a]	548[abc]	495[abc]	288[cd]	597[ab]	490[abc]	155[d]	329[bcd]	(0)	(0)	270·4	234·2
Muguga 1970 ..	2,510[a]	2,342[ab]	2,348[ab]	1,733[c]	1,981[bc]	1,859[c]	1,256[d]	1,114[d]	(11)	—	444·4	362·7
Hybrid 512												
Muguga 1970	3,112[a]	2,638[ab]	3,067[a]	2,802[ab]	3,278*[a]	2,380[ab]	2,522[ab]	2,022[b]	70	—	417	850
Kitale 1970	6,371[a]	6,256[ab]	5,524[bc]	5,097[cd]	5,656[abc]	5,250[cd]	4,959[cd]	4,344[d]	234	—	971·7	793·3
Hybrid 611.*C*												
Kitale 1970 . ..	6,264[a]	5,982[abc]	6,388*[ab]	4,581[c]	4,473[c]	4,926[bc]	4,560[c]	2,159	(38)	—	1,562	1,275
SORGHUM												
Serena												
Serere 1970	3,302[a]	3,190[ab]	3,212[ab]	3,308*[a]	3,240[a]	2,678[bc]	2,475[c]	2,288[c]	605[d]	260[d]	544·3	471·2

the latter were not used in the analyses except in the Hybrid 512 experiments. In other treatments, however, fairly large yields were obtained, which were not significantly different (at the 5 per cent level) from the control in many cases, ranging from two to six of the treatments in different experiments. The least severe A treatments produced the least crop loss; after treatment A.1 the yield was never significantly different from the control, and only in one experiment after A.2; after A.3, however, it was significantly less in four out of five cases, and ranged from 47 to 90 per cent of the control.

After the somewhat more severe B treatments (100 per cent leaf removal), yields were not significantly less than in the control in three cases with B.1, two with B.2 and one with B.3; in the other cases yields ranged from 82 per cent of the control right down to 25 per cent.

After C.1 treatments, yields were always significantly smaller than the control, and ranged from 68 down to 34 per cent of it.

It is clear from the results that with each of treatments A, B and C, the later it was carried out the greater was the reduction in yield in most cases. Averaging the yields expressed as percentages of the controls, for the five maize trials (Table VIII), produces the following figures:

	Treatment 1	*Treatment* 2	*Treatment* 3
A	92	92	72
B	88	78	61
C	53	8	0

Only in the case of A.1 and A.2 does the latter treatment not produce a lower figure. This linear effect shows a meaningful trend even when individual figures may not be statistically different from one another. There is, in A and B, a marked drop resulting from the third treatments, for which the yields in B treatments are mostly considerably lower than in A; these last treatments were applied close to the time when cobs begin to form, and the more complete leaf removal in B treatments has a more drastic effect on the growth and development of the cob. Furthermore, silking plants have the largest proportion of young leaves, since there is a rapid opening of leaves just before and during the time the tassels are emerging. These young leaf tissues on older plants are more vigorous and effective in the synthesis of nutrients (Eldredge, 1935), and are more responsible than the older leaves for the development of the grain (Allison and Watson 1966). The complete leaf removal at stage 3 in B treatments therefore has a relatively more drastic effect than the 75 per cent removal at the same stage in A treatments. However, the greater leaf removal in B treatments also produces considerably more crop loss than in A treatments even in younger stages, for while in all except one instance there was greater loss (considerable in some cases) after B.2 than after B.1 treatments, there was no corresponding difference between A.1 and A.2, the latter even being followed by larger yields in three cases out of five.

The nature of the drastic effect of C.2 and C.3 treatments will be discussed later.

Regarding differences between varieties, a clear difference appears between the early maturing Katumani and late-maturing H.512 at Muguga in 1970. The latter was relatively less affected and gave yields not significantly different from the controls after a much wider range of treatments. There were no such clear differences between H.512 and H.611C at Kitale, but the difference in time of development was much less. In H.512, grown at both places in 1970, yields were, perhaps, slightly less affected by defoliation at Muguga than at Kitale, but the difference was not great.

In the case of sorghum, the effect of treatments on yield was in general less than in maize. Averaging the yields, expressed as percentages of the controls, for all nine treatments (Table VII) produces a figure of 71 per cent, the highest corresponding figure for a maize experiment being 69 and the lowest 52 (the latter admittedly that carried out at Muguga in 1969 under unsatisfactory conditions). More realistic is to compare the average figures for the A and B treatments only, being more within the normal range of armyworm damage; that for sorghum is 91 per cent; the corresponding figures for maize (1970 experiments only) range from 76 to 89 per cent, averaging 83 per cent. An interesting feature of the sorghum experiment is that even after C.3 treatment a measurable crop was obtained, so that all nine treatments could be subjected to statistical analysis; this is the result of tillering following stem removal, which will be discussed later.

Number of cobs or heads per plot, and weight of grain per cob or head. The results of these two factors potentially capable of directly affecting yield are given in Table II.

Plate 2—Simulated armyworm damage to maize

A. Treatment C.1, stem cut down to ground level; showing regeneration 11 days after treatment.

B. Plots at Muguga, 1969, with treatments C.1 (background) and C.2 (foreground), respectively 57 and 33 days after treatment; in the former all plants have survived, but few in the latter.

C. General view of plots after second series of treatments, Muguga, 1969.

D. Actual armyworm damage to maize, Arusha, 1968.

TABLE II—ARTIFICIAL DAMAGE EXPERIMENT: MEAN VALUES, FOR CONTROLS AND TREATMENTS, FOR THE NUMBER OF COBS OR HEADS PER PLOT AND WEIGHT OF GRAIN PER COB OR HEAD

**Healthy or "usable" cobs or heads only; **all cobs, including "undeveloped" or diseased ones. Other details as in Table I.*

Experiment	No. of cobs or heads per plot / wt. of grain per cob or head (g.)	Control mean	Treatment means									Least significant difference (5 per cent level)	
			A.1	A.2	A.3	B.1	B.2	B.3	C.1	C.2	C.3	Between treatment means	Between control and treatment means
MAIZE													
Katumani Composite "B" Muguga 1969	cobs/plot**	$28{\cdot}4^{a}$	$25{\cdot}2^{ab}$	$25{\cdot}8^{ab}$	$28{\cdot}0^{a}$	$25{\cdot}8^{ab}$	$27{\cdot}0^{ab}$	18 2	$23{\cdot}0^{b}$	(3)	—	4·03	3·49
	grain/cob**	$21{\cdot}1^{a}$	21 6^{a}	$19{\cdot}1^{ab}$	10 3^{bc}	$22{\cdot}6^{a}$	17 5^{ab}	$8{\cdot}4^{c}$	$14{\cdot}2^{abc}$	—	—	8·80	7·62
Muguga 1970	cobs/plot**	35 7^{ab}	36 0^{a}	$34{\cdot}2^{ab}$	$34{\cdot}2^{ab}$	$34{\cdot}0^{ab}$	$35{\cdot}8^{ab}$	$32{\cdot}0^{b}$	26·8	(0·25)	—	3·99	3·25
	grain/cob*	$71{\cdot}1^{a}$	$65{\cdot}6^{ab}$	$71{\cdot}1^{a}$	$52{\cdot}1^{cd}$	$59{\cdot}0^{bc}$	$52{\cdot}8^{cd}$	$39{\cdot}6^{c}$	43 8^{dc}	—	—	11·01	8·99
Hybrid 512 Muguga 1970	cobs/plot**	$37{\cdot}2^{a}$	$33{\cdot}0^{ab}$	$37{\cdot}8^{a}$	$35{\cdot}5^{a}$	$36{\cdot}0^{a}$	$32{\cdot}0^{ab}$	$33{\cdot}5^{ab}$	$26{\cdot}5^{b}$	(1·25)	—	7·39	6·03
	grain/cob*	$82{\cdot}5^{a}$	$78{\cdot}8^{a}$	$83{\cdot}4^{a}$	$80{\cdot}0^{a}$	$90{\cdot}4^{a}$	76 6^{a}	$77{\cdot}6^{a}$	$76{\cdot}3^{a}$	—	—	—	—
Kitale 1970	cobs/plot**	$35{\cdot}4^{a}$	34 0^{a}	$34{\cdot}2^{a}$	$32{\cdot}5^{a}$	$34{\cdot}2^{a}$	$36{\cdot}0^{a}$	$34{\cdot}2^{a}$	$28{\cdot}5^{a}$	—	—	—	—
	grain/cob*	192^{a}	191^{ab}	171^{cd}	173^{bcd}	183^{abc}	160^{d}	157^{d}	158^{d}	—	—	18·40	15·00
Hybrid 611.C Kitale 1970	cobs/plot**	$31{\cdot}9^{a}$	$30{\cdot}0^{ab}$	$30{\cdot}8^{ab}$	$25{\cdot}2^{b}$	$26{\cdot}5^{ab}$	$25{\cdot}2^{b}$	$28{\cdot}0^{ab}$	16·2	—	—	6·79	5·54
	grain/cob*	203^{abc}	217^{a}	216^{ab}	190^{cd}	178^{d}	192^{bcd}	172^{d}	134	—	—	24·10	19·68
SORGHUM Serere 1970	heads/plot*	69^{ab}	70^{ab}	69^{ab}	76^{a}	64^{ab}	64^{ab}	58^{b}	59^{b}	26^{c}	20^{c}	14·1	12·2
	grain/head*	480^{a}	459^{ab}	465^{ab}	437^{ab}	508^{a}	416^{ab}	472^{ab}	391^{b}	25^{c}	13^{c}	101·5	87 9

(*a*) *Number of cobs.* Of all the 30 A and B treatments of maize, in only three cases was the number of cobs per plot significantly less than in the control: in treatment B.3 for Katumani at Muguga in 1969, and in A.3 and B.2 for Hybrid 611C at Kitale in 1970. These could very well be due to chance; although two of them were late treatments which might be expected to affect cob formation more than earlier ones, the B.2 result at Kitale was followed by a B.3 treatment producing a smaller loss than it and even than the corresponding B.1 treatment. Four out of five C.1 treatments, however, were significantly less than in the control, and so removal of the stem even at an early stage would appear to reduce the number of cobs.

With sorghum the effect of treatments is no greater than in maize; all A and B treatments and also C.1 did not produce a significant reduction of the number of heads.

(*b*) *Weight of grain per cob or head.* Here the case is clearly different. Out of all 30 A and B maize treatments, 12 produced a weight of grain significantly less than in the control. Of these, two were in the first treatments, three in the second, and seven in the third. This would appear to be due to the loss of leaf area at the third treatment adversely affecting the growth and development of the cobs rather than their initial formation; the C.1 yield per cob was slightly bigger than the B.3, although the number of cobs was much smaller.

Averaging the treatment results expressed as percentages of the controls (Table VII) for all A and B treatmens, and C.1, we get the following figures:

	A.1	A.2	A.3	B.1	B.2	B.3	C.1
No. cobs/plot..	94	97	92	93	93	86	72
Wt. grain/cob..	99	97	81	97	86	71	74

This clarifies the contrasting effects of treatments with regard to these two parameters, in that when the treatment is applied later, the reduction in quantity of grain per cob (especially in treatments B.2 and B.3) is appreciably greater than that in the number of cobs per plot.

In the case of sorghum the result of treatment on yield of grain per head was very much less, except in the case of C treatments where the second and third took very drastic effect, much more so than on the number of heads per plot. The reason for this was a simple one: owing to the retarded development, the heads did not reach the stage susceptible to shoot-fly damage until after the insecticide treatment had ceased to be protective; it was therefore not due directly to the treatment itself.

TABLE III—DATA ON THE NUMBER AND DEVELOPMENT OF COBS IN KATUMANI MAIZE AFTER THE DIFFERENT TREATMENTS AT MUGUGA, 1969

Treatment	Number of cobs per plot	Per cent of plants with no cobs	Per cent of cobs undeveloped (no grain formed)
Control ..	28	3	39
A.1 .	25	6	36
A.2 ..	26	10	35
A.3 .	27	7	64
B.1 ..	26	8	37
B.2 ..	27	5	36
B.3 ..	18	35	75
C.1 ..	23	19	52
C.2 ..	3	61	100
C.3 ..	All	plants	died

Plants without cobs, and with undeveloped cobs. Records were kept, as stated earlier, only in the 1969 maize experiment at Muguga, when growing conditions (especially rainfall) were poor fertilizer was not used, and the treatments therefore had accentuated effects due to reduced resistance and regeneration. Results are given in Table III, together with the number of cobs per plot for comparison.

Defoliation at the third stage affected the development of cobs much more than that at the two earlier stages. This is particularly evident in the reduction of the number of cobs in B.3 treatments and of the development of grain in both B.3 and A.3. The more severe effect of B.3 than A.3 treatments may be related to the fact that it involves complete leaf removal at a time when photosynthetic activity is particularly important for the growth and maturation of the cobs. A.3 treatment resulted in no increase in the percentage of plant without cobs, but in considerable decrease in formation of grain, presumably because the 25 per cent of leaf area remaining was sufficient to enable cobs to form, but not enough for the grain to develop. Complete absence of leaves at this

TABLE IV—AVERAGE HEIGHT OF FULL-GROWN MAIZE PLANTS (IN CM. MEASURED TO THE BASE OF THE TOPMOST LEAF), IN EXPERIMENTS WITH MAIZE AT MUGUGA, 1969 AND 1970

*Values in each experiment with the same letter appended are not significantly different from one another. (N.R.—not recorded, N.S.—not significant; *not included in significance tests)*

Treatment	Control mean	Treatment means									Least significant difference (5 per cent level)	
		A.1	A.2	A.3	B.1	B.2	B.3	C.1	C.2	C.3	Between treatment means	Between control and treatment means
Katumani Composite "B"												
1969	95[a]	97[a]	92[a]	85[a]	93[a]	82[a]	68[a]	77[a]	(31)*	—	N.S.	N.S.
1970	138[a]	133[ab]	139[a]	118[c]	120[bc]	115[c]	116[c]	113[c]	N.R.	—	14·9	12·2
Hybrid 512												
1970	240[a]	234[ab]	235[ab]	217[abc]	241[ab]	214[bc]	197[c]	200[c]	N.R.	—	29·0	23·7

TABLE V—THE PERIOD BETWEEN PLANTING AND 50 PER CENT TASSELING, 50 PER CENT SILKING AND 50 PER CENT COB FORMATION IN CERTAIN OF THE MAIZE EXPERIMENTS IN 1970

Other data as in Table 3.

Experiment	Period (days)	Control mean	Treatment means									Least significant difference (5 per cent level)	
			A.1	A.2	A.3	B.1	B.2	B.3	C.1	C.2	C.3	Between treatment means	Between control and treatment means
MUGUGA													
Katumani Composite "B"	Planting date to: 50 per cent silking	86[a]	86[a]	87[a]	87[a]	89[a]	88[a]	90[a]	101	—	—	5·5	4·5
	50 per cent cob formation	84[a]	84[ab]	85[ab]	85[ab]	86[ab]	85[ab]	88[b]	94	—	—	4·34	3·54
Hybrid 512	50 per cent tasseling	104[a]	107[a]	103[a]	106[a]	106[a]	104[a]	105[a]	112[a]	—	—	4·89	3·99
	50 per cent silking	116[a]	123[bc]	116[a]	117[ab]	118[ab]	118[ab]	118[ab]	129[c]	—	—	6·7	5·5
	50 per cent cob formation	110[a]	116[b]	111[ab]	115[ab]	112[ab]	112[ab]	112[ab]	123	—	—	6·6	5·4
KITALE													
Hybrid 512	50 per cent tasseling	80[a]	81[ab]	81[ab]	80[ab]	80[ab]	83[b]	79[a]	91	96	—	3·16	2·58
	50 per cent silking	86[a]	89[a]	88[a]	88[a]	90[a]	89[a]	88[a]	97	106	—	4·32	3·53
Hybrid 611 C	50 per cent tasseling	94[a]	94[a]	94[a]	94[a]	98[b]	95[ab]	93[a]	105	—	—	3·09	2·52
	50 per cent silking	105[a]	105[a]	107[a]	108[a]	112[a]	111[a]	108[a]	136	—	—	9·33	7·61

time (B.3) resulted not only in still further reduction in development of grain (75 per cent of cobs "undeveloped"), but also in considerable reduction of cobs (35 per cent of plants without any). Corresponding C treatments, as would be expected, had more severe effects on all results.

Height of maize plants

Average measurements for the maize experiments at Muguga (the only ones in which they were taken) are given in Table IV. The difference between the 1969 and 1970 results for Katumani maize is an indication of the difference in growing conditions in the two years. Nearly all plants were very small in 1969, but were so variable that there was no significant difference between all A and B treatments and C.1. The difference between all A treatments combined and all B treatments was not quite statistically significant at the 5 per cent level. There was, however, a significant linear trend over the three A treatments and over the three B treatments, in that the later the damage occurred the shorter were the plants.

The same was not true under the better conditions and more even growth of 1970, when only two treatments for Katumani and four for Hybrid 512 were not significantly different from the controls. The reduction in height was more marked after B treatments than after A, but the linear effect was clear only in the case of B treatments for Hybrid 512. After C.1 treatments plants still managed to reach a good height although the initial stem was cut right down.

Rate of development of maize

Since the time taken for development of the reproductive organs could be important in determining the maturation of the crop under conditions of limited rainfall, records of the period from planting to 50 per cent tasselling, 50 per cent silking and 50 per cent cob formation were taken for several of the maize experiments in 1970 (Table V).

The results can be seen to be very even on the whole, with few significant differences between treatments except in the case of C treatments, which caused a marked delay in development even in the one case where a C.1 result was not significantly different from the control. The difference between A and B treatments and controls is rarely sufficient to be of much importance; the anomalous figures for A.1 with Hybrid 512 at Muguga are likely to be a freak result, not due simply to the treatment.

Number of tillers (in sorghum)

It would seem likely that cutting back leaves and more especially the stem in early stages of growth would increase tillering of sorghum plants. In fact, there was no significant difference, in the average number of stems (including fertile and infertile ones) per plant, either between treatments or between any treatment and the control; the figure for the latter was exactly equal to the average for those of all the treatments. The average figures were as follows, in descending order:—

C.1	A.3	C.2	B.2	B.3	Control	A.1	B.1	A.2	C.3
3·11	3·02	3·00	2·96	2·89	2·82	2·78	2·72	2·65	2·29

TABLE VI—NUMBER OF PLANTS SURVIVING IN ALL PLOTS GIVEN C TREATMENTS, INCLUDING ALL FOUR ROWS IN EACH PLOT

(*Estimated figure based on two centre rows only; N.R. not recorded).

Experiment	Original number of plants	C.1		C.2		C.3	
		No.	Per cent survival	No.	Per cent survival	No.	Per cent survival
MAIZE							
Katumani Composite "B"							
Muguga 1969	208	204	98	33	16	0	0
Muguga 1970	288	230*	80	11	4	—	—
Hybrid 512							
Muguga 1970	288	280*	97	39	13	—	—
Kitale 1970	288	N.R.	—	24	8	—	—
Hybrid 611.*C.*							
Kitale 1970	288	N.R.	—	10	3	—	—
SORGHUM							
Serere 1970	256	175	68	109	43	100	39

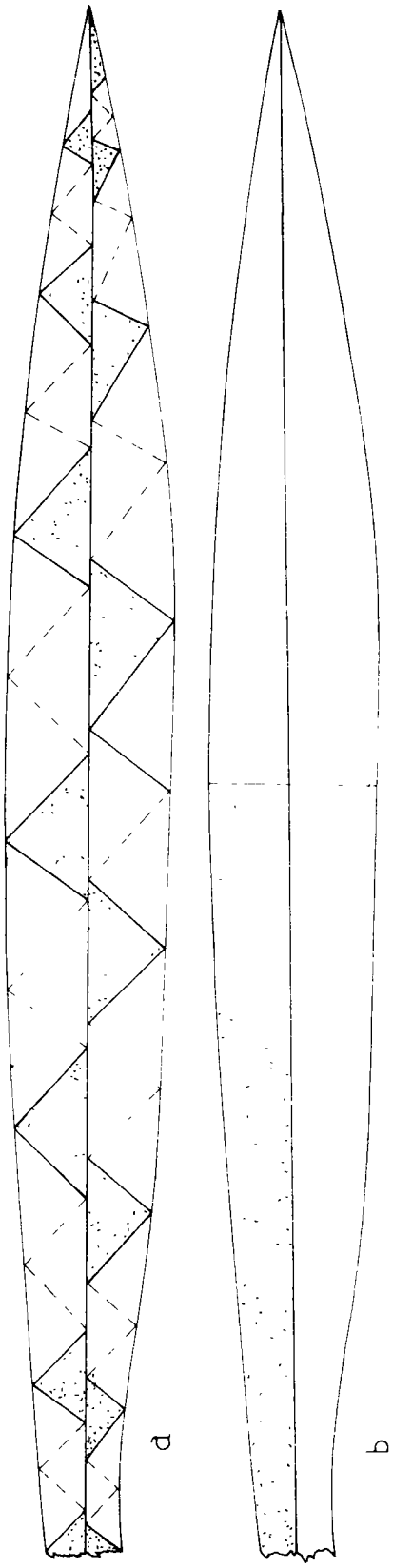

Fig. 1—Two methods of removing 75 per cent of the leaf area. The parts of the leaves left on the plant are stippled

The most interesting result is an expected one, that the highest number resulted from cutting the stem down to ground level at the earliest treatment (C.1).

The number of fertile stems ("heads", Table II) per plant was somewhat higher (1.17) in the control, than in C.2 (0.96) and C.3 (0.8), so that any increase in number of tillers per plant by early C treatments was offset by a decrease in the number of fertile ones.

Number of plants surviving after "C" treatment

This was of special interest in relation to C treatments in maize, owing to the interesting behaviour of the growing point described below. The total number surviving in all four rows in all plots together, expressed also as a percentage of the total original number of plants, are given in Table VI.

In maize there were no survivors after treatment C.3 at Muguga in 1969, and the treatment was not carried out in maize experiments in 1970. After C.2 the percentage survival ranged from three to 16 only; after C.1, however, the survival rate was much higher, up to nearly 100 per cent in two cases.

In the case of sorghum, although the percentage survival was never very high even after C.1, a considerable number of plants (39 per cent) survived after treatment C.3. This is related to the fact that tillering, which does not occur in maize, takes place low down on the plant and the growing points of the stems are very close to ground level.

Comparison of the two methods of effecting treatment A

The two methods (*a*) and (*b*) of removing 75 per cent of the leaf area are illustrated in Figure 1. Method (*a*) was used exclusively at Muguga in 1969, and method (*b*) at Kitale and Serere in 1970; at Muguga in 1970 at the first round of treatments, four of the eight plots concerned (two of each variety of maize) were treated by method (*a*) and four by method (*b*), while at all later treatments method (*b*) only was used. Since regeneration after early A treatments is very considerable in any case, this was hardly a fair test of any difference there might be. There was some indication that method (*a*) was slightly more damaging than (*b*), especially for Katumani maize, but it was not possible to do significance tests and it is unlikely that any differences were sufficient to affect the overall results.

Hanway (1969) used six methods of effecting 50 per cent defoliation. None resembled method (*a*); four of them involved removing whole leaves in different combinations, but two consisted of removing either the distal or the lateral half of each leaf; he found no significant differences in the effects of all his methods.

Dungan (1930) also used a number of methods for 50 per cent defoliation. These included methods essentially identical with (*a*) and (*b*), in the one case removing 4 in. segments of leaf blade alternately along the two sides of the midrib, in the other the distal half of the leaf. Crop loss resulting from these methods were 41.66 per cent and 43.49 per cent respectively, a difference considered to be of minimal importance, especially since the treatment was made at the "silking" stage. However, he found that removal of one side of each leaf produced a loss of only 29.66 per cent, and removal of half of the leaves a loss of 22.46 per cent; he deduced from all these results that the distal half of the leaf is nutritionally more important.

Behaviour of the growing point in maize and its relation to the effects of "C" treatments

Measurements were made, for three different varieties of maize, for the position of the growing point in relation to ground level, at various stages of growth of the plant from the five leaf stage to that when nine leaves had opened and part of the 10th. For any one variety and stage of growth the figures were very variable. Even when averaged for several plants and plotted on a graph, the curve produced was somewhat irregular; assessments were therefore smoothed to a three stage running mean, and Figure 2 is the result of plotting these figures on a graph.

It can be seen that in two of the varieties (Katumani Composite "B" and Hybrid 512), the growing point rises to ground level at about the eight-leaf stage; in the third variety (Hybrid 611.C) it does not do so until somewhat later, between the 8½ and 9-leaf stage. These should be regarded as average results, subject to considerable variation. Hanway (1963) found that the growing point reached ground level earlier, at the 4-leaf stage, and that by the 8-leaf stage it was 6-8 in. above it. This may have been due to a different variety of maize, or possibly to different planting depths

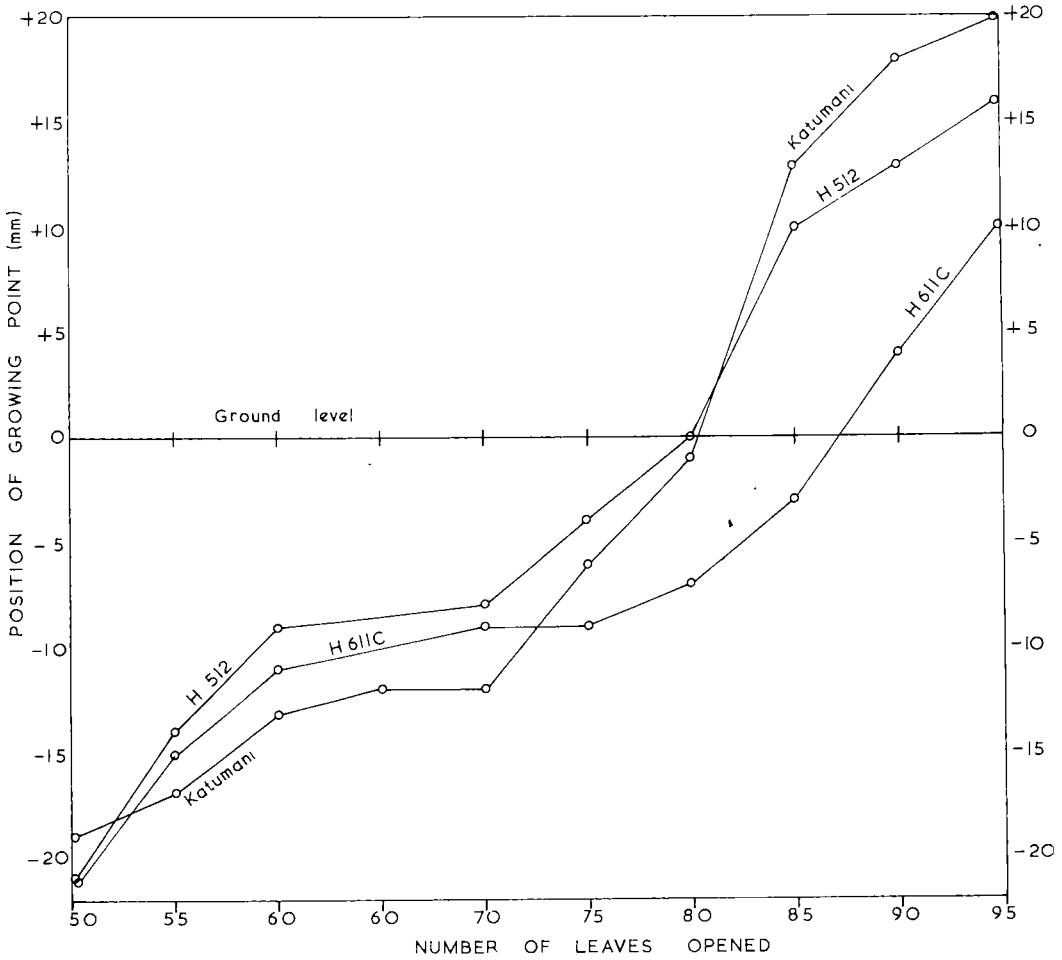

Fig. 2—The position of the growing point in relation to ground level, in plants with different numbers of leaves opened, in three varieties of maize, Katumani Composite "B", and hybrids 512 and 611.C. Distances below ground level are marked with a minus, and above ground level with a plus sign

If the stem is cut right down before the growing point has reached ground level, the meristem is not destroyed and the plant is likely to survive, although with reduced vigour; if growing point has risen above ground level at the time of cutting, it will be destroyed and the whole plant with it. For this reason plants nearly all survived after C.1 treatment and all perished after C.3. C.2 treatment was so timed that the growing point was approximately at ground level at the time of cutting, so that a varying proportion of the plants survived. One would expect a lower survival of Katumani maize in 1969, when treatments were carried out a little later, and a higher one in Hybrid 611.C since the growing point remains below ground level and is protected a little longer; no explanation is offered for the fact that neither of these things happened. However, the experiments demonstrate the general principal that in maize there is a critical stage of growth after which severe stem damage results in complete destruction of the plant.

General comments on results

The major results of all treatments in all experiments are summarized in Table VII in which treatment results are expressed as percentages of the controls in each case. All those results which are not significantly different

TABLE VII—COMPARISON OF EFFECTS OF TREATMENTS ON YIELD AND VARIOUS OTHER ASPECTS OF DEVELOPMENT OF THE PLANTS, WITH EACH OTHER AND WITH THE CONTROLS

Figures are rounded off to the nearest whole number. An asterisk (*) signifies the treatment effect is not significantly different from the control at the 5 per cent level, figures in brackets were not included in the significance tests.

Parameter and Experiment		Mean value for all control plots	Mean values for all treatments, expressed as a percentage of the control								
			A.1	A.2	A.3	B.1	B.2	B.3	C.1	C.2	C.3
YIELD											
Maize	Katumani, Muguga 1969	615 gm.	89*	80*	47	97*	80*	25	53	(0)	(0)
	Katumani, Muguga 1970	2,510 gm.	93*	94*	69	79	74	50	44	(0·4)	—
	Hybrid 512, Muguga 1970	3,112 gm.	85*	99*	90*	105*	76*	81*	65	2	—
	Hybrid 512 Kitale 1970	6,371 gm.	98*	87	80	89*	82	78	68	37	—
	Hybrid 611.C Kitale 1970	6,264 gm.	95	102*	73	71	79	73	34	(0·6)	—
Sorghum	Serena, Serere 1970	3,302 gm.	97*	97*	100*	98*	81	75	69	18	8
HEIGHT OF PLANTS											
Maize	Katumani, Muguga 1969	95 cm.	102*	97*	89*	98*	86*	72*	81*	(33)	—
	Katumani, Muguga 1970	138 cm.	96*	101*	86	87	83	84	82	N.R.	—
	Hybrid 512, Muguga 1970	240 cm.	97*	98*	90*	100*	89	82	83	N.R.	—
NO. OF COBS OR HEADS/PLOT											
Maize	Katumani, Muguga 1969	28	89*	93*	100*	93*	96*	64	82	(11)	—
	Katumani, Muguga 1970	36	100*	94*	94*	94*	100*	89*	75	(0·7)	—
	Hybrid 512, Muguga 1970	37	89*	103*	96*	97*	86*	91*	72	(3·4)	—
	Hybrid 512, Kitale 1970	35	96*	97*	92*	96*	102*	96*	80*	—	—
	Hybrid 611.C, Kitale 1970	32	94*	97*	79	83*	79	88*	51	—	—
Sorghum	Serena, Serere 1970	69	101*	100*	110*	93*	93*	84*	86*	38	29
WEIGHT OF GRAIN/COB OR HEAD											
Maize	Katumani, Muguga 1969	21 gm.	102*	91*	49	107*	83*	40	67*	—	—
	Katumani, Muguga 1970	71 gm.	92*	100*	73	83	74	56	62	—	—
	Hybrid 512, Muguga 1970	82 gm.	96*	101*	97*	110*	93*	94*	92*	—	—
	Hybrid 512, Kitale 1970	192 gm.	99*	89	90	95*	83	82	82	—	—
	Hybrid 611.C, Kitale 1970	203 gm.	107*	106*	94*	88	95*	85	66	—	—
Sorghum	Serena, Serere 1970	480 gm.	96*	97*	91*	106*	87*	98*	81	5	3
TIME TO 50 PER CENT TASSELLING											
Maize	Hybrid 512, Muguga 1970	104 days	103*	99*	102*	102*	100*	101*	108*	—	—
	Hybrid 512, Kitale 1970	80 days	101*	101*	100*	100*	104	99*	114	120	—
	Hybrid 611.C, Kitale 1970	94 days	100*	100*	100*	104*	101*	99*	112	—	—
TIME TO 50 PER CENT SILKING											
Maize	Katumani, Muguga 1970	86 days	100*	101*	101*	103*	102*	105*	117	—	—
	Hybrid 512, Muguga 1970	116 days	106	100*	101*	102*	102*	102*	111	—	—
	Hybrid 512, Kitale 1970	86 days	103*	102*	102*	105*	103*	102*	113	123	—
	Hybrid 611.C, Kitale 1970	105 days	100*	102*	103*	107*	106*	103*	130	—	—
TIME TO 50 PER CENT COB FORMATION											
Maize	Katumani, Muguga 1970	84 days	100*	101*	101*	102*	101*	105	112	—	—
	Hybrid 512, Muguga 1970	110 days	105	101*	105*	102*	102*	102*	112	—	—
No. of occasions (out of 30) when treatment values did not differ significantly from control			28	28	21	24	20	17	6	0	0

from the controls are indicated, and amount to a surprisingly large total in the case of several of the treatments.

A.1 and A.2 treatments each failed to produce significant differences from controls in no less than 28 out of the 30 results; B.1 came next with 24, and all A and B treatments together showed no significant difference from controls in over half of the results. B treatments all produced more significant differences from the control than the corresponding A treatments, as is to be expected, and results of C treatments were nearly always significantly different.

DISCUSSION

Accounts in the literature of earlier defoliation experiments on maize have been referred to before. Frequently the object has been to simulate hail damage, and is not unconnected with insurance policies against this catastrophe. Defoliation has often been carried out in later stages of growth of the plants than we are concerned with here. Hanway for instance (1969) did not perform the first treatments (50 per cent and 100 per cent leaf removal) until the 10th leaf had appeared. Crop loss under these circumstances was always significant; for an early maturing variety it was 19 per cent even with only 50 per cent defoliation at the earliest (10-leaf) stage, and was most severe at tasselling when 100 per cent defoliation resulted in virtually complete loss of crop.

It is, however, unusual to find armyworm attack in the later stages of growth, and few farmers are unduly concerned about this pest when their maize has reached a height of two to three feet. Defoliation usually occurs at a younger stage than this, and the plant has more time to recuperate.

A study of Table VII will show that the treatments, all of which are severe in terms of percentage defoliation, have a surprisingly small effect upon yield, and upon other features of the plant, such as height and duration of development, which might be expected to affect yield indirectly. When leaves or parts of leaves only are removed and the stems left intact, subsequent growth of the plant and crop are significantly affected only in a minority of cases. This is especially true when the treatments are applied early in the growth of the plant (treatments A.1, A.2, B.1 and to a lesser extent B.2). The regeneration of the plants after such treatments is well illustrated by photographs (Plate 1); the growth of new leaves some days after treatment is little impaired, or not at all and they replace functionally the older treated leaves, which would in any case die off by the time four or five new leaves are formed. It is therefore clear that in earlier stages plants can withstand considerable defoliation without being seriously retarded.

Even when the stem is removed at the first treatment (C.1), a considerable amount of regeneration occurs (Plate 2. A and B), new leaves growing out of the stump from the growing point which is still underground and therefore undamaged; however, in this case only approximately 50 per cent of the normal crop is ultimately realized.

It should be borne in mind that insect damage in the field may produce effects different from or additional to those resulting from an equivalent amount of artificial defoliation (Jones, Dunning and Humphies, 1955). In the present experiments, for instance, a factor which is not accounted for is the sustained attack by armyworm over a period of time, as distinct from experimental defoliation on one occasion only. New leaves are eaten off as soon as they appear, and the plant is deprived of foliage and photosynthetic activity for perhaps several days. Environmental factors such as rainfall are even more important for effective regeneration under these circumstances.

It would be useful to be able to forecast the ultimate amount of leaf damage from the density of young larvae, at the stage when a decision should be taken whether it is economical to control them; but the problem in the field is complex, owing to unknown factors such as subsequent mortality of larvae and conditions governing regeneration of the plant, notably rainfall. A few facts are known, however, which provide pointers; Brown and Odiyo (1968) provided laboratory data from which it can be calculated. for instance, that in a maize plant with six to seven opened leaves, all the leaf area could be destroyed by less than two larvae during their life-time, although the continuing growth of the plant during this period is a complication. Laboratory experiments have also shown that larvae in sufficient numbers can eat a plant up to at least the six leaf stage right down to the ground, stem and all; this could result in 50 per cent crop loss even if all the plants survive. There is a need for

the systematic collection of field data on the range of defoliation resulting from known densities of larvae, from which probablities of economic damage justifying control can be assessed with reasonable accuracy even if only in a majority of cases and not in all. The experimental results described in the present work could then be put to more extensive practical use. There is need also for repetition of the experiments described here for other crops, especially rice, wheat and finger millet (*Eleusine*).

It is appropriate to mention briefly also certain other agricultural effects of armyworm activity in the field, relating to both cultivated crops and grass pastures; even if irrelevant to the particular title of this paper, these comments serve to draw attntion to related problems deserving of further research.

Reports are quoted by Hattingh (1941) of armyworm attack adversely affecting grazing by destroying good pasture grasses and leaving less palatable ones, and after severe attack following "veld burns" some farmers have claimed that pastures may take two or three years before they are suitable for grazing again. On the other hand, many farmers in East Africa state that, given adequate rain, grasses grow better after an outbreak, and in northern Uganda farmers are stated to give preference to areas which have been attacked, for planting crops.

In large numbers, larvae produce a significant amount of faeces. A partial analysis at Muguga of a small quantity of dried armyworm droppings gave the following values (as percentages of total weight), compared with similar published figures for farmyard manure:—

	Nitrogen	Phosphorus	Potassium	Calcium	Magnesium
Armyworm droppings ..	0·21	0·02	0·30	0·36	0·10
Farmyard manure	0·44	0·11	0·27	0·30	0·10

Although the amounts of nitrogen and phosphorus are small compared with farmyard manure, the nutrients might be expected to have some effect as a fertilizer.

As a test to find out whether droppings have any obviously harmful effect on maize growth, a small experiment was set up at Muguga with 10 maize plants, each planted in a separate bucket of about 20 litres capacity in potting soil (a forest soil mixed with coffee hulls, stone chips, boma manure, and N, P and K added). The plants were exposed to the rain, and were watered occasionally when the soil became dry. At the four to five leaf stage, and on seven further occasions at three day intervals a layer of armyworm droppings was spread over the soil surface in five of the 10 buckets; the rain washed the droppings down into the soil. At weekly intervals records were made for each plant of its height, the number of open leaves, the circumference of the stem just above the second leaf, and of the length and width of the biggest leaf.

Up to 39 days after the first treatment there was little difference in most of these features between treated and untreated plants, but by the 32nd day treated plants appeared much more healthy and robust, with thicker stems and bigger leaves. Although the quantity of droppings used was excessive, equivalent to some 1,000 larvae per square yard, it is clear that there was no harmful, but rather a favourable effect resulting from their application, and in fact it is difficult to imagine any way in which larvae could have a harmful effect upon plant growth, either in cultivated crops or in pastures.

Another possible side effect, this time of a harmful nature, relates to the deaths sometimes reported in cattle feeding in pastures attacked by armyworm. This is attributed to direct stomach action following ingestion. For example, on a farm near Nairobi in 1957, 50 cows were reported to have died at the time of a heavy armyworm invasion, and in 1971 eight cows died out of a herd of 50 on the same farm under similar circumstances. The animals were reported as appearing "bloated", and staggering before collapsing and dying. Dr. M. Stevens, of the Veterinary Research Laboratory, Kabete, made a post-mortem examination, and kindly provided a full clinical account of the symptoms (personal communication); without giving details here, it may be stated that while there were extensive effects upon the tongue, trachea, lymph nodes, liver, udder, gut lining and eyes (corneal opacities), no evidence was found of any of the usual fly or tick-borne diseases, virus infection, or poisoning by

arsenic or insecticidal compounds commonly used, and no obvious poisonous plants were found in the area; this recalls symptoms described by Dimes (1938) in cattle which were said to have ingested many armyworms on fresh growth of grass with rains after drought: the animals "became unsteady in their gait, their eyesight appeared impaired and they stumbled about the bush until they eventually fell to the ground. Many cattle were lying about—some dead, some in *extremis*—while others were recovering".

While it thus appears highly probable that such deaths may be attributable to the effect of ingestion of armyworms, no clear proof of such cause and effect has so far been produced so far as we are aware. Pending further research, however, it is nevertheless advisable where possible to keep stock off pasture infested with armyworms, as Masai herdsmen are reported to do, presumably as the result of bitter experience.

It is thus clear that there is room for further research on a variety of side effects of armyworm activity and that the simple effects of defoliation on crop loss alone do not provide a complete picture.

Summary

An account is given of experiments with maize and sorghum, in Kenya and Uganda respectively, in which varying amounts of defoliation were related to resulting crop loss and to other effects upon the growth of the plants. The defoliation was intended to simulate damage by the armyworm *Spodoptera exempta;* the objective was to obtain data from which one might predict the extent of crop loss to be expected from varying intensities of armyworm attack, and thus assess the advisability of carrying out control with economic advantage.

The experiments showed that, especially under favourable environmental conditions, regeneration after defoliation of both maize and sorghum in the earliest stages of growth treated can be virtually complete. In general, it is concluded that, when defoliation affects leaves only and takes place very early in the growth of the plant (up to the five-leaf stage in maize) crop loss is negligible; it becomes more important the later the stage of growth at which it occurs, and when the stem is also damaged crop loss is always serious, and complete when damage occurs at later growth stages. In sorghum, due to tillering, the effects on crop loss of defoliating the primary stem are always less important; destruction of the primary stem itself results in crop loss which is severe, although less so than in the corresponding maize treatment, and a crop, although greatly reduced, is produced even when it is destroyed at the latest stage of treatment.

The curious behaviour of the primary meristem in maize, and its effect on the results of destruction of the stem at different stages, is described.

Differences between defoliation as effected experimentally, and as brought about by armyworm attack, are discussed. Sustained attack and destruction of foliage by armyworm may ultimately render their damage more serious.

Certain side effects of armyworm, such as the fertilizer—effect of their droppings, and their possible poisoning action when ingested by cattle in pastures, are also discussed.

Acknowledgements

Our thanks are expressed to Dr. L. H. Penny and Mr. E. W. Mwenda of the Maize Genetics Research Division of E.A.A.F.R.O., and Dr. Dean Barry of the Sorghum Research Division of E.A.A.F.R.O., for initiating and carrying out experiments with maize at Kitale and sorghum at Serere respectively; and to Mr. M. N. Harrison, formerly of the same Maize Genetics Research Division, and Mr. P. T. Walker of the Tropical Pesticides Research Unit, Porton Down, for advice on the conduct of the experiments. Dr. J. B. D. Robinson, of the Chemistry Division, E.A.A.F.R.O., kindly undertook the chemical analysis of armyworm droppings for the major elements.

We are grateful to members of the staff of the Armyworm Division, E.A.A.F.R.O., Muguga, for their relentless work involved in recording data at Muguga.

Special thanks are due to Miss Sheila Green, of the Centre for Overseas Pest Research, London, for statistical analysis of the results and advice in treatment of the data.

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REVIEW

NATIVE TREES OF CANADA by R. C. Hosie, published by Queen's Printer for Canada, Ottawa, 1969, pp. 380. Price Can. $5.00 (paper-bound), Can. $8.00 (cloth-bound).

This book is the seventh edition of the original volume, first published in 1917. There is a brief, but comprehensive introduction describing Canadian forests and forest regions (with accompanying maps) and a discussion on tree terminology of value to the layman. The main section of the book provides a detailed treatment of 140 indigenous species, arranged by botanical families. "Native" trees are defined as single-stemmed, perennial woody plants growing to a height of more than ten feet (approx. 3 m.), and which are indigenous to Canada. Exotic species capable of growing in Canada are occasionally mentioned in the text for comparison with indigenous trees. Appendices at the back of the book contain sections on botanical authorities associated with Canadian tree names, meanings of tree names, a short bibliography of other North American tree books, and keys to Canadian species.

Every edition of this book has notably improved on its predecessor, and the current offering is no exception. Distribution maps, photographs, and illustrations are of a very high standard and reproduced on good quality paper. Important features of each species are well illustrated, and the descriptive text on form, habitat, botanical characteristics, and economic importance is both comprehensive and non-technical. Dichotomous keys are so well arranged and simple that any reader, with assistance from the introductory notes, should easily be able to identify any tree.

Anyone with an interest in trees in different parts of the world should acquire this book. Though of limited practical value to potential users on the African content, *Native Trees of Canada* is recommended for inclusion in libraries at all botanical, agricultural, or forestry institutions.

H.D.G.